

Store at: +2+8° C.

Presentation:

Cod. SU007-SP CONT: R 2 x 50 mL + Cal 1 x 5 mL.

Cod. SU007 CONT: R 2 x 125 mL + Cal 1 x 5 mL.

Cod. SU007-B CONT: R 8 x 125 mL + Cal 1 x 5 mL.

Procedure

Quantitative determination of calcium.

Only for in vitro use in clinical laboratory (IVD)

TEST SUMMARY

Calcium with Arsenazo III (1,8-Dihydroxy-3,6-disulpho-2,7-naphthalene-bis (azo)-dibenzeneazonic acid), at neutral pH, yields a blue colored complex. The intensity of the colour formed is proportional to the calcium concentration in the sample^{1,2,3}.

REAGENTS COMPOSITION

R (Arsenazo)	Imidazol Buffer pH 6.5 Arsenazo III	100 mmol/L 120 mmol/L
Calcium Cal	Calcium aqueous primary calibrator	10 mg/dL

PRECAUTIONS

R: H360- May damage fertility or the unborn child.

CAL: H290-May be corrosive to metals.

Follow the precautionary statements given in MSDS and label of the product.

REAGENT PREPARATION AND STABILITY

All the reagents (R.1) (Calcium cal) are ready to use.

Calcium Cal: Proceed carefully with this product because due its nature it can get contaminated easily.

Signs of Reagent deterioration:

- Presence of particles and turbidity.
- Blank absorbance (A) at 650 nm $\geq$ 0.50

All the components of the kit are stable until the expiration date on the label when stored tightly closed at 2-8° C, protected from light and contaminations prevented during their use. Do not use reagents over the expiration date.

SPECIMEN

Serum or plasma¹: Separated from cells as rapidly as possible. Blood anticoagulants with oxalate or EDTA are not acceptable since these chemicals will strongly chelate calcium.

Urine¹: Collect 24 hour urine specimen in calcium free containers. The collecting bottles should contain 10 ml of diluted Nitric acid (50% v/v). Record the volume. Dilute a sample 1/2 in distilled water. Mix. Multiply results by 2 (dilution factor).

Stability of the samples: Calcium is stable 10 days at 2-8° C.

MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 650 nm.
- Matched cuvettes 1.0 cm. light path.

General laboratory equipment (note 1),

TEST PROCEDURE

- Assay Conditions
 - Wavelength: 650 nm.
 - Cuvette: 1 cm light path.
 - Temperature 37°C /15-25°C.
- Adjust the instrument to zero with distilled water.
- Pipette into a cuvette:

	Blank	Standard	Sample
R 1 (mL)	1.0	1.0	1.0
Standard (Note 2, 3) (μL)	--	10	--
Sample (μL)	--	--	10

- Mix and incubate for 2 minutes.
- Read the absorbance (A) of the samples and calibrator, against the Blank. The colour is stable for at least 1 hour.

CALCULATIONS

Serum or plasma:

$$\text{Calcium (mg/dL)} = \frac{(A) \text{ Sample} - (A) \text{ Blank}}{(A) \text{ Standard} - (A) \text{ Blank}} \times 10 \text{ (Cal. conc.)}$$

Urine 24:

$$\text{Calcium (mg/24 h)} = \frac{(A) \text{ Sample} - (A) \text{ Blank}}{(A) \text{ Standard} - (A) \text{ Blank}} \times 10 \times \text{vol. (dL) urine/24 h}$$

Conversion factor: mg/dL x 0.25 = mmol/L.

QUALITY CONTROL

Control sera are recommended to monitor the performance of the procedure, Control H Normal Ref. QC003 and Control H Pathological Ref. QC004. If control values are found outside the defined range, check the instrument, reagents and calibrator for problems.

Serum controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions.

REFERENCE VALUES¹

Serum or plasma:

Adults 8.5-10.5 mg /dL $\approx$ 2.1-2.6 mmol/L

Children 10 -12 mg/dL $\approx$ 2.5 - 3 mmol/L

Newborns 8 -13 mg/dL $\approx$ 2 - 3.25 mmol/L

Urine:

Adults 50 - 300 mg/24h $\approx$ 1.25 - 7.5 mmol/24h

Children 80 -160 mg/24h $\approx$ 2 - 4 mmol/24h

(These values are for orientation purpose).

It is suggested that each laboratory establish its own reference range.

CLINICAL SIGNIFICANCE

Calcium is the most abundant and one of the most important minerals in the human body. Approximately 99% of body calcium is found in bones.

A decrease in albumin level causes a decrease in serum calcium. Low levels of calcium are found in hypoparathyroidism, pseudohypoparathyroidism, vitamin D deficiency, malnutrition and intestinal malabsorption. Among causes of hypercalcemia are cancers, large intake of vitamin D, enhanced renal retention, osteoporosis, sarcoidosis, thyrotoxicosis, hyperparathyroidism^{1,6,7}.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

REAGENT PERFORMANCE

Measuring Range:

From detection limit of 0.026 mg/dL. to linearity limit of 32 mg/dL., under the described assay conditions.

If results obtained were greater than linearity limit, dilute the sample 1/2 with NaCl 9 g/L. and multiply result by 2.

Precision:

	Intra-assay n= 20		Inter-assay n= 20	
Mean (mg/dL)	8.35	14.28	8.58	14.57
SD	0.08	0.08	0.19	0.34
CV %	0.95	0.59	2.24	2.31

- Sensitivity: 1 mg/dL. = 0.0316 A.

Accuracy:

Results obtained GPL reagents did not show systematic differences when compared with other commercial reagents.

The results obtained using 50 samples were the following:

Correlation coefficient (r)²: 0.9506

Regression Equation : y= 0.8944x + 1.3421

The results of the performance characteristics depend on the analyzer used.

INTERFERING SUBSTANCES

- No interferences were observed with triglycerides up to 1.25 g/L^{1,2,3}.
- A list of drugs and other interfering substances with calcium determination has been reported by Young et. al^{2,3}.

NOTES

- It is recommended to use disposable material. If glassware is used the material should be scrupulously cleaned with diluted 1/2 HNO₃ in water and then thoroughly rinsed it with distilled water.
- Most of the detergents and water softening products used in the laboratories contain chelating agents. A defective rinsing will invalidate the procedure.
- Calibration with the aqueous standard may cause a systematic error in automatic procedures. In these cases, it is recommended to use a serum Calibrator.
- Use clean disposable pipette tips for its dispensation.

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